What happens to the camera?

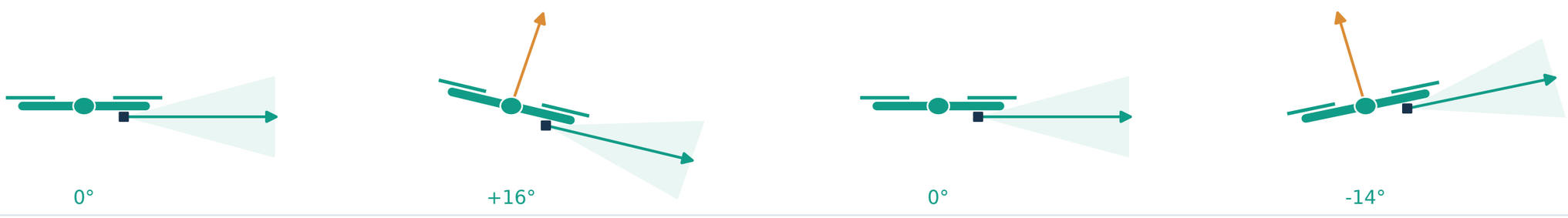
A side-view example of forward acceleration, cruising, and braking

Shaded cone + colored arrow: camera view • Orange arrow: thrust direction

1 HOVER	2 ACCELERATE	3 CRUISE	4 BRAKE
Level; no forward acceleration	Tilt forward; camera looks downward	Near level if drag is small	Tilt back; camera looks upward

Custom quadrotor

Rigid-body quadrotor



Both methods can produce this tilt sequence.

Custom: approximate rate response. Rigid body: torque + inertia determine the response and any tracking error.

Illustration only: angles, spacing and viewing cones are schematic; these are not simulation results or rendered 3DGS RGB.
Camera optical axis assumed along body +X here. The real mounting rotation changes the absolute viewing direction; drag can require tilt during cruise.